

EXPERIMENT – 8

Privilege Escalation Using TryHackMe Platform

Aim

To understand and demonstrate Linux privilege escalation techniques by exploiting system misconfigurations and vulnerabilities using the TryHackMe platform.

Tools / Platform Required

Web Browser (Chrome / Firefox)

Internet Connection

TryHackMe Account (Already Created)

TryHackMe AttackBox (Built-in Linux Environment)

Linux Privilege Escalation Tools (LinPEAS, sudo, find, etc.)

Theory (Brief)

Privilege Escalation is a technique used to gain higher-level permissions than originally assigned.

In Linux systems, attackers often exploit misconfigured permissions, vulnerable binaries, or weak sudo rules to escalate from a normal user to root.

TryHackMe provides a legal and controlled environment to practice privilege escalation techniques and understand real-world attack scenarios.

Steps to Perform the Experiment

Step 1: Login to TryHackMe

1. Open browser and go to: <https://tryhackme.com>
2. Login using your registered credentials.

Step 2: Open Privilege Escalation Room

1. In the search bar, type **Linux Privilege Escalation**.
2. Click on the appropriate room.
3. Click **Join Room**.

Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Initial Enumeration

1. Check current user:
2. `whoami`
3. Check system information:
4. `uname -a`

5. Check sudo permissions:
6. `sudo -l`

Step 5: Identify Privilege Escalation Vectors

Common techniques explored:

- Sudo misconfigurations
- SUID binaries
- Weak file permissions
- Environment variable abuse
- Cron job misconfigurations

Example command:

```
find / -perm -4000 2>/dev/null
```

Step 6: Exploit Vulnerability

1. Execute the vulnerable binary or misconfigured command.
2. Escalate privileges to root user.
3. Verify escalation:
4. `whoami`

Step 7: Capture Flags / Complete Tasks

1. Obtain root-level access.
2. Capture the required flags.
3. Mark all tasks as completed.

Output

```
root@ip-10-49-137-12: ~
File Edit View Search Terminal Help
root@ip-10-49-137-12:~# whoami
root
root@ip-10-49-137-12:~# uname -a
Linux ip-10-49-137-12 6.17.0-1012-aws #12~24.04.1-Ubuntu SMP Mon Apr 6 17:36:28
UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
root@ip-10-49-137-12:~#
```

```
root@ip-10-49-137-12:~# sudo -l
Matching Defaults entries for root on ip-10-49-137-12:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin,
    use_pty

User root may run the following commands on ip-10-49-137-12:
    (ALL : ALL) ALL
root@ip-10-49-137-12:~#
```

```
User root may run the following commands on ip-10-49-137-12:
    (ALL : ALL) ALL
root@ip-10-49-137-12:~# find / -perm -4000 2>/dev/null

/usr/lib/polkit-1/polkit-agent-helper-1
/usr/lib/x86_64-linux-gnu/dcv/dcvpamhelper
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/openssh/ssh-keysign
/usr/lib/xorg/Xorg.wrap
```

```
/usr/bin/su
/usr/bin/sudo
/usr/bin/chsh
/usr/bin/chfn
/usr/bin/newgrp
/usr/sbin/mount.nfs
/usr/sbin/pppd
/opt/BurpSuiteCommunity/burpbrowser/145.0.7632.45/chrome-sandbox
/snap/core24/1643/usr/bin/chfn
/snap/core24/1643/usr/bin/chsh
/snap/core24/1643/usr/bin/gpasswd
/snap/core24/1643/usr/bin/mount
/snap/core24/1643/usr/bin/newgrp
/snap/core24/1643/usr/bin/passwd
/snap/core24/1643/usr/bin/su
/snap/core24/1643/usr/bin/sudo
/snap/core24/1643/usr/bin/umount
/snap/core24/1643/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/snap/core24/1643/usr/lib/openssh/ssh-keysign
/snap/core24/1643/usr/lib/polkit-1/polkit-agent-helper-1
/snap/core22/2411/usr/bin/chfn
/snap/core22/2411/usr/bin/chsh
/snap/core22/2411/usr/bin/gpasswd
/snap/core22/2411/usr/bin/mount
/snap/core22/2411/usr/bin/newgrp
/snap/core22/2411/usr/bin/passwd
/snap/core22/2411/usr/bin/su
/snap/core22/2411/usr/bin/sudo
/snap/core22/2411/usr/bin/umount
/snap/core22/2411/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/snap/core22/2411/usr/lib/openssh/ssh-keysign
/snap/core22/2411/usr/libexec/polkit-agent-helper-1
root@ip-10-49-137-12:~#
root@ip-10-49-137-12:~#
root@ip-10-49-137-12:~#
root@ip-10-49-137-12:~#
root@ip-10-49-137-12:~# whoami
root
root@ip-10-49-137-12:~#
```

Successful privilege escalation from normal user to root.

System commands executed with elevated privileges.

Result

Thus, privilege escalation was successfully performed using the TryHackMe platform.

The experiment provided hands-on understanding of Linux privilege escalation techniques and system security weaknesses.

Important Note

This experiment is performed **strictly for educational and ethical purposes** using the TryHackMe learning platform.